SG No. 208 O^2 $P4_232$ [cubic]

* plus set: +[0, 0, 0]

Table 1: Wyckoff site: 2a, site symmetry: 23.

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: 4b, site symmetry: .32

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,5,9,14,19,24]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,7,12,13,17,21]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,8,10,15,20,22]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4,6,11,16,18,23]

Table 3: Wyckoff site: 4c, site symmetry: .32

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[1,5,9,14,19,24]
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[2,7,12,13,17,21]
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[3,8,10,15,20,22]
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[4,6,11,16,18,23]

Table 4: Wyckoff site: 6d, site symmetry: 222..

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,3,4]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[5,6,7,8]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[9,10,11,12]
4	$\left[0, \frac{1}{2}, 0\right]$	[13,14,15,16]
5	$\left[\frac{1}{2}, 0, 0\right]$	[17,18,19,20]
6	$\left[0, 0, \frac{1}{2}\right]$	[21,22,23,24]

Table 5: Wyckoff site: 6e, site symmetry: 2.22

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[1,4,18,19]
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[2,3,17,20]
3	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[5,8,14,15]
4	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[6,7,13,16]
5	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[9,12,21,24]
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[10,11,22,23]

Table 6: Wyckoff site: `6f`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,4,18,19]`
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,3,17,20]`
3	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,8,14,15]`
4	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,13,16]`
5	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,12,21,24]`
6	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,11,22,23]`

Table 7: Wyckoff site: `8g`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,9]`
2	$[-x, -x, x]$	`[2,7,12]`
3	$[-x, x, -x]$	`[3,8,10]`
4	$[x, -x, -x]$	`[4,6,11]`
5	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-x\right]$	`[13,17,21]`
6	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[14,19,24]`
7	$\left[x+\frac{1}{2}, \frac{1}{2}-x, x+\frac{1}{2}\right]$	`[15,20,22]`
8	$\left[\frac{1}{2}-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[16,18,23]`

Table 8: Wyckoff site: `12h`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4]`
2	$[-x, 0, 0]$	`[2,3]`
3	$[0, x, 0]$	`[5,8]`
4	$[0, -x, 0]$	`[6,7]`
5	$[0, 0, x]$	`[9,12]`
6	$[0, 0, -x]$	`[10,11]`
7	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	`[13,16]`
8	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	`[14,15]`
9	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,20]`
10	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	`[18,19]`
11	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	`[21,24]`
12	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	`[22,23]`

Table 9: Wyckoff site: `12i`, site symmetry: `2..`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,4]`
2	$\left[-x, 0, \frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{2}, x, 0\right]$	`[5,8]`
4	$\left[\frac{1}{2}, -x, 0\right]$	`[6,7]`
5	$\left[0, \frac{1}{2}, x\right]$	`[9,12]`

continued ...

Table 9

No.	position	mapping
6	$\left[0,\ \frac{1}{2},\ -x\right]$	[10,11]
7	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[13,16]
8	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[14,15]
9	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[17,20]
10	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[18,19]
11	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	[21,24]
12	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	[22,23]

Table 10: Wyckoff site: 12j, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,4]
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	[2,3]
3	$\left[0,\ x,\ \frac{1}{2}\right]$	[5,8]
4	$\left[0,\ -x,\ \frac{1}{2}\right]$	[6,7]
5	$\left[\frac{1}{2},\ 0,\ x\right]$	[9,12]
6	$\left[\frac{1}{2},\ 0,\ -x\right]$	[10,11]
7	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[13,16]
8	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[14,15]
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[17,20]
10	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[18,19]
11	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	[21,24]
12	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	[22,23]

Table 11: Wyckoff site: 12k, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$	[1,19]
2	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$	[2,17]
3	$\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$	[3,20]
4	$\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$	[4,18]
5	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$	[5,14]
6	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$	[6,16]
7	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$	[7,13]
8	$\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$	[8,15]
9	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[9,24]
10	$\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[10,22]
11	$\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[11,23]
12	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[12,21]

Table 12: Wyckoff site: `12l`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{4}, y, y+\frac{1}{2}\right]$	`[1,18]`
2	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	`[2,20]`
3	$\left[\frac{3}{4}, y, \frac{1}{2}-y\right]$	`[3,17]`
4	$\left[\frac{1}{4}, -y, \frac{1}{2}-y\right]$	`[4,19]`
5	$\left[y+\frac{1}{2}, \frac{1}{4}, y\right]$	`[5,15]`
6	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	`[6,13]`
7	$\left[\frac{1}{2}-y, \frac{3}{4}, y\right]$	`[7,16]`
8	$\left[\frac{1}{2}-y, \frac{1}{4}, -y\right]$	`[8,14]`
9	$\left[y, y+\frac{1}{2}, \frac{1}{4}\right]$	`[9,21]`
10	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	`[10,23]`
11	$\left[y, \frac{1}{2}-y, \frac{3}{4}\right]$	`[11,22]`
12	$\left[-y, \frac{1}{2}-y, \frac{1}{4}\right]$	`[12,24]`

Table 13: Wyckoff site: `24m`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[-x, -y, z]$	`[2]`
3	$[-x, y, -z]$	`[3]`
4	$[x, -y, -z]$	`[4]`
5	$[z, x, y]$	`[5]`
6	$[z, -x, -y]$	`[6]`
7	$[-z, -x, y]$	`[7]`
8	$[-z, x, -y]$	`[8]`
9	$[y, z, x]$	`[9]`
10	$[-y, z, -x]$	`[10]`
11	$[y, -z, -x]$	`[11]`
12	$[-y, -z, x]$	`[12]`
13	$\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}-z\right]$	`[14]`
15	$\left[y+\frac{1}{2}, \frac{1}{2}-x, z+\frac{1}{2}\right]$	`[15]`
16	$\left[\frac{1}{2}-y, x+\frac{1}{2}, z+\frac{1}{2}\right]$	`[16]`
17	$\left[x+\frac{1}{2}, z+\frac{1}{2}, \frac{1}{2}-y\right]$	`[17]`
18	$\left[\frac{1}{2}-x, z+\frac{1}{2}, y+\frac{1}{2}\right]$	`[18]`
19	$\left[\frac{1}{2}-x, \frac{1}{2}-z, \frac{1}{2}-y\right]$	`[19]`
20	$\left[x+\frac{1}{2}, \frac{1}{2}-z, y+\frac{1}{2}\right]$	`[20]`
21	$\left[z+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-x\right]$	`[21]`
22	$\left[z+\frac{1}{2}, \frac{1}{2}-y, x+\frac{1}{2}\right]$	`[22]`
23	$\left[\frac{1}{2}-z, y+\frac{1}{2}, x+\frac{1}{2}\right]$	`[23]`
24	$\left[\frac{1}{2}-z, \frac{1}{2}-y, \frac{1}{2}-x\right]$	`[24]`